A Fourier view of the Riemann hypothesis

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Abstract

An elementary proof of the Riemann hypothesis is provided using Fourier analysis on the alternating zeta function *η(s)*. The real series $|\eta(s)|^2$, whose zeros are equivalent to the non-trivial zeros, increases along $\Im(s)$in the left half of the critical strip $0<\Re(s)<1/2$.

1. Introduction

The Riemann zeta function is intimately connected to the distribution of prime numbers through the Euler product:

(1.1) $$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_p \frac{1}{1-p^{-s}}$$

where *p* runs over all prime numbers and *s* is a complex variable. The zeta function is extended to the complex plane by analytical continuation, except for its simple pole at s=1. Its logarithmic derivative can be expressed as the Dirichlet series:

(1.2) $$\varphi(s) = -\frac{\zeta'(s)}{\zeta(s)} = \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^s}$$

where Λ(n) denotes the von Mangoldt function, which is nonzero precisely when n is a prime power. In contrast, the prime zeta function *P(s)* is defined directly in terms of the primes:

(1.3) $$P(s) = \sum_{prime\ p} \frac{1}{p^s}$$

For $0<\Re(s)<1$, the Zeta-function can be evaluated through the alternating zeta function *η(s)*, according to:

(1.4) $$\eta(s) = (1-2^{1-s})\zeta(s)$$

This representation provides a convergent formulation in the critical strip. The zeta function is also related to ζ(1-s) through its functional equation:

(1.5) $$\zeta(s) = 2^s \pi^{s-1} \sin(\frac{\pi s}{2}) \Gamma(1-s) \zeta(1-s)$$

The non-trivial zeros of the zeta-function, which lie in the critical strip, $0<\Re(s)<1$, are believed to be deeply connected to the distribution of prime numbers. The Riemann hypothesis, first formulated by Bernhard Riemann in 1859[1], states that all non-trivial zeros lie on the critical line $\Re(s)=1/2$. Despite extensive efforts to prove or disprove the hypothesis, it remains one of the most important open problems in mathematics [2–6].

2. Theorem

Lemma 2.1 The Riemann zeta function represents the Fourier transform of a one-dimensional weighted logarithmic series of all natural numbers.

Proof:

Let *f(x)* represent a series of point mass at position set $\{x_i = ln(n_i)\}$ with weight set $\{w_i = n_i^{-\beta}\}$ in one dimensional (1D) direct space **x** (β is a tuning parameter to adjust the weight):

(2.1) $$f(x) = \sum_{n=1}^{\infty} n^{-\beta} \delta(x - \ln(n))$$

where δ is the Dirac delta function.

The Fourier transform of *f(x)* is:

$$\zeta(k) = \int_{-\infty}^{+\infty} \sum_{n=1}^{\infty} n^{-\beta} \delta(x - \ln(n)) e^{-2\pi i k x} dx = \sum_{n=1}^{\infty} n^{-\beta} \int_{-\infty}^{+\infty} \delta(x - \ln(n)) e^{-2\pi i k x} dx$$

(2.2) $$= \sum_{n=1}^{\infty} n^{-\beta} e^{-2\pi i k \ln(n)} = \sum_{n=1}^{\infty} n^{-\beta} n^{-2\pi i k}$$

$$= \sum_{n=1}^{\infty} \frac{1}{n^s}$$

where $s = \beta + 2\pi k i$ is a complex number as in (1.1), and *k* is the frequency variable in 1D *k*-space.

Property 2.1 Similarly, the alternating zeta function η(k) represents the same series as ζ(k) but with different weight:

(2.3) $\eta(k) = \sum_{n=1}^{\infty} (-1)^{n-1} n^{-s} = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^{\beta}} e^{-2\pi i k \ln(n)}$

Also the Dirichlet series $\varphi(s)$ (1.2) represents the Fourier transform of the logarithmic series of the prime powers $\{x_i=ln(q_i)\}$ with weight $\{w_i=\Lambda(q_i)q_i^{-\beta},\}$ and the prime zeta function $P(k)$ (1.3) represents the Fourier transform of the logarithmic series of all primes $\{x_i=ln(p_i)\}$with weight $\{w_i=p_i^{-\beta}\}$. Since $\eta(k)$ represents $\zeta(k)$ in the critical strip, we'll mainly use it in the following discussion.

Property 2.2 The frequency component set $\{k_i\}$ of $\zeta(s)$ in k-space is determined by the position set $\{x_i\}$ and is independent of the weighting parameter β. It is expected that $\zeta(k)$, $\eta(k)$, $P(k)$ and $\varphi(k)$ all share the same component set $\{k_i\}$ because they are all logarithmic series of the natural numbers and the logarithm of every natural number is a finite linear combination of the logarithms of the primes. Figure 1 shows 4 calculated Fourier transforms. It can be seen that $\zeta(k)$ and $\eta(k)$ have minima at the non-trivial zero positions marked by green dotted lines with $\{\gamma_i = 2\pi k_i = 14.13, 21.02, 25.01, \ldots\}$, where $\varphi(k)$ and $P(k)$ have maxima at these positions. Therefore, the imaginary parts of the non-trivial zeros are the frequency components of $\zeta(k)$ and their positions can be determined from the minima of ζ(k) and η(k), or the maxima of $P(k)$ and $\varphi(k)$ with different accuracy.

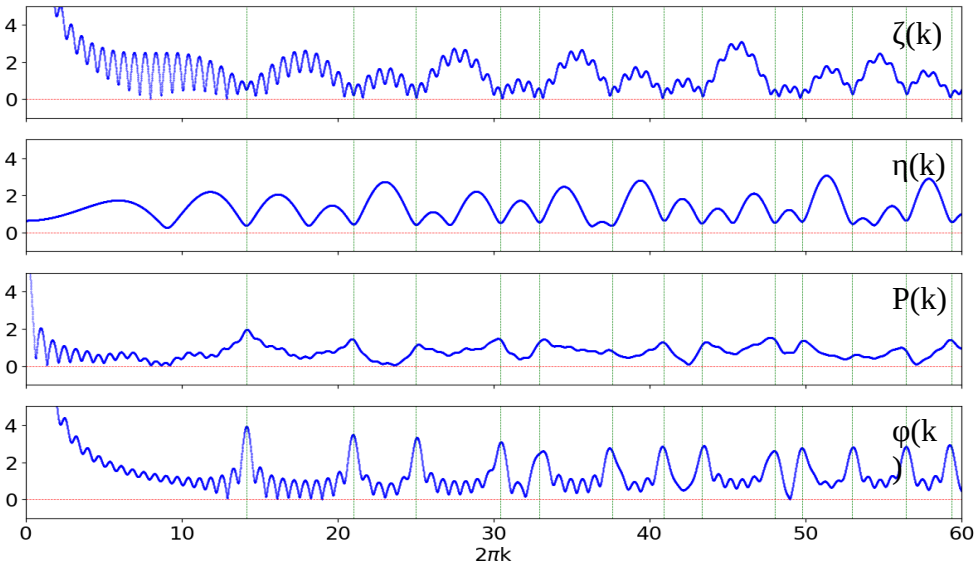


Figure 1. Fourier transforms of four logarithmic series of numbers with β=0.75. The frequency positions of the non-trivial zeros are marked with green lines. Vertical axis: amplitude of the spectrum.

Property 2.3 The weight set $\{w_i\}$ modulates the appearance or amplitudes of $\{k_i\}$ in k-space through the tuning parameter β. η(k) at a different β represents a different view of the same component set $\{k_i\}$; therefore, the Fourier transform can be expressed as $\eta(k)|_\beta$ to reflect the role of β. For convenience, we use η(s) in the following proof. The variation of β spans the 1D k-space into a complex plane with k running along the vertical imaginary axis and β on the real axis.

Theorem 2.1 $\eta(s) = \sum_{k=1}^{\infty} \frac{(s)_k}{2^{s+k}\,k!}\zeta(s+k)$

Proof:

The difference d_n between the odd-even pairs $(2n-1)^{-s}$, $(2n)^{-s}$ of η(s) (2.3) is

$$d_n = (2n-1)^{-s} - (2n)^{-s} = -(2n)^{-s}\left(1-\left(1-\frac{1}{2n}\right)^{-s}\right)$$

$$= -\frac{1}{(2n)^s}\left(1-\sum_{k=0}^{\infty}\frac{(s)_k}{k!}\frac{1}{(2n)^k}\right) = \sum_{k=1}^{\infty}\frac{(s)_k}{k!}\frac{1}{(2n)^{s+k}} \tag{2.4}$$

where binomial expansion is used and $(s)_k = s(s+1)...(s+k-1)$ with $(s)_0 = 1$. Then,

$$\eta(s) = \sum_{n=1}^{n=\infty} d_n = \sum_{k=1}^{\infty}\frac{(s)_k}{2^{s+k}\,k!}\zeta(s+k) \tag{2.5}$$

Theorem 2.2 Real series $|\eta(s)|^2$ increases with $\Im(s)$ in the left half of the critical strip $0<\Re(s)<1/2$.

Proof :

The intensity of η($s=\beta+i\gamma$) can be expressed as:

$$I(s) = |\eta(s)|^2$$

$$= \sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}\frac{(-1)^{n_1+n_2}}{(n_1 n_2)^{\beta}}\cos\left(\gamma\ln\left(\frac{n_2}{n_1}\right)\right) \tag{2.6}$$

$$= \sum_{n=1}^{\infty}\frac{1}{n^{2\beta}} + 2\sum_{n_1=1}^{\infty}\sum_{n_2=n_1+1}^{\infty}\frac{(-1)^{n_1+n_2}}{(n_1 n_2)^{\beta}}\cos\left(\gamma\ln\left(\frac{n_2}{n_1}\right)\right) \tag{2.7}$$

$$= I_e - I_o \tag{2.8}$$

where I_o is the sum of odd terms (n_1+n_2 is odd) and I_e is the sum of even terms including the diagonal sum in (2.7). Since I(s) is non-negative, $I_e>=I_o$.

A non-trivial zero occurs if and only if *I(s)=0*, so the even sum should equal to the odd sum:

(2.9) $$\lim_{N\to\infty}\sum_{k=1}^{2k<N}\sum_{n=1}^{N-2k}\frac{\cos\left(\gamma\ln\left(\frac{n+2k}{n}\right)\right)}{(n(n+2k))^{\beta}}+\frac{1}{2}\sum_{n=1}^{N}\frac{1}{n^{2\beta}}=\lim_{N\to\infty}\sum_{k=1}^{2k-1<N}\sum_{n=1}^{N-2k+1}\frac{\cos\left(\gamma\ln\left(\frac{n+2k-1}{n}\right)\right)}{(n(n+2k-1))^{\beta}}$$

Since the frequency components $\{\gamma_i\}$ exist for all β values in the critical strip (0,1/2] and they can be determined independent of β, this constrain is mainly on β and it determines the optimal β value that turns the normal minima at positions $\{\gamma_i\}$ to zeros.

Figure 2 shows 6 *I(s)* spectra at β=0.0, 0.1, 0.2, 0.3, 0.4 and 0.5. It is apparent that for β<1/2, the gap $\Delta=I_e-I_o$ increases with γ and attenuates with β, and it oscillates in between two envelopes.

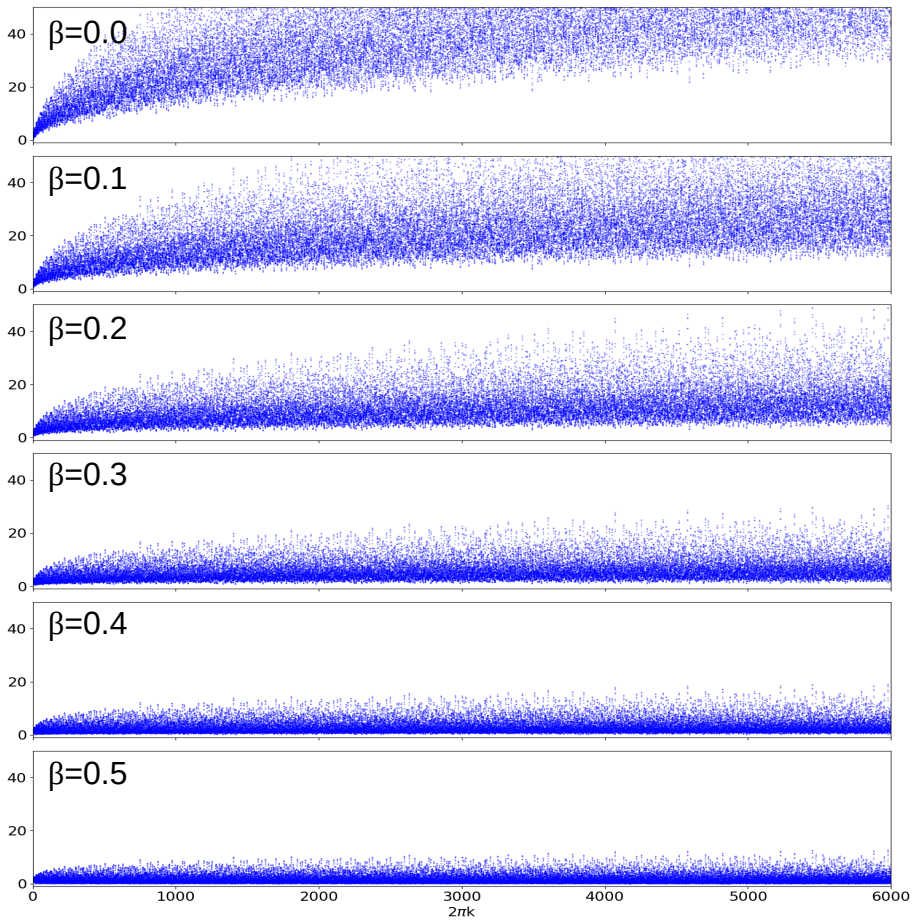


Figure 2. Amplitude spectra of η(s=β+iγ) show the increase of I_e-I_o along imaginary axis (γ=2πk) and attenuation with β. Vertical axis: amplitude √I(s). The upper and lower envelopes of the spectra both appear linear in I(s).

We also calculated spectra with β close to 0.5. The mean density of the spectra (partial sum with N=500,000) for β (0, 0.6) in three intervals of γ are listed in the following table.

γ \ β	**0.0**	**0.1**	**0.2**	**0.3**	**0.4**	**0.49**	**0.5**	**0.51**	**0.6**
[2000, 2100]	812.43	277.71	99.01	37.75	15.96	8.31	7.79	7.31	4.49
[4000,4100]	1623.24	482.45	149.77	50.01	18.8	9.02	8.4	7.84	4.6
[6000,6100]	2457.72	683.01	199.29	63	22.62	10.49	9.73	9.04	5.16

Both envelopes appear linear in the intensity spectra I(s). The lower envelope can be estimated from the minima of the intensity spectra in each interval Δγ=10 of γ from 10. Assuming linearity and zero intercept, the estimated slopes are listed below and the lowest is $8.3405e^{-7}$ at β=0.5. For every γ,

β	**0.0**	**0.1**	**0.2**	**0.3**	**0.4**	**0.49**	**0.5**	**0.51**	**0.6**
slope	0.1298	0.021	$2.7e^{-3}$	$2.47e^{-4}$	$1.01e^{-5}$	$8.59e^{-7}$	$8.34e^{-7}$	$8.56e^{-7}$	$2.81e^{-6}$

the intensity at β=0.5 is the lowest, so there is no zero detected for β<1/2 for γ values up to 6100.

The above direct evidence perfectly supports the Riemann hypothesis and is a result of Theorem 2.1. Each term of η(s) in (2.5) generates a spectrum of rising background along $\Im(s)=\gamma$:

$$\eta(s) = \frac{\beta+i\gamma}{2^{s+1}} [\zeta(s+1) + \frac{s+1}{4}\zeta(s+2) + \frac{(s+1)(s+2)}{24}\zeta(s+3) + ...] \tag{2.10}$$

For example, the first term ζ(s+1) is convergent and has no zeros for $0<\beta=\Re(s)$ [2], the intensity of ζ(s+1) at all γ positions including the non-zero minima is magnified by $\sim\gamma^2$:

$$r_1 = |\frac{\beta+i\gamma}{2^{s+1}}|^2 \approx \frac{\gamma^2}{2^{2\beta+2}} \tag{2.11}$$

which results in a rising background. In factor, term ζ(s+k) is convergent and zero-free for 1-k<β. Non-trivial zeros occur only when all these spectra become coherent for a specific β. Therefore, I(s) has a generally increasing background along γ. For a given β ϵ(0, ½), as long as η(s) is zero-free in a lower range of γ, for example, $10< \gamma<=\gamma_0 =100$ (which covers the ~ 30 frequency components), it will be zero-free forward for $\gamma>\gamma_0$.

Because of the increase of I_e-I_o background along γ for 0<β<½, and considering the fact that there is no zero (I_e-I_o>0) below γ=(6000, 6100), we can conclude that there is no non-trivial zero forward for γ>6100 in the left half of the critical strip 0<β<1/2; and according to the functional equation (1.5), no zero can exist in the other half of the critical strip $1/2<\Re(s)<1$. Therefore, the non-trivial zeros have to lie on the critical line $\Re(s)=1/2$. Figure 3 shows how precisely the non-trivial zeros land on the critical line.

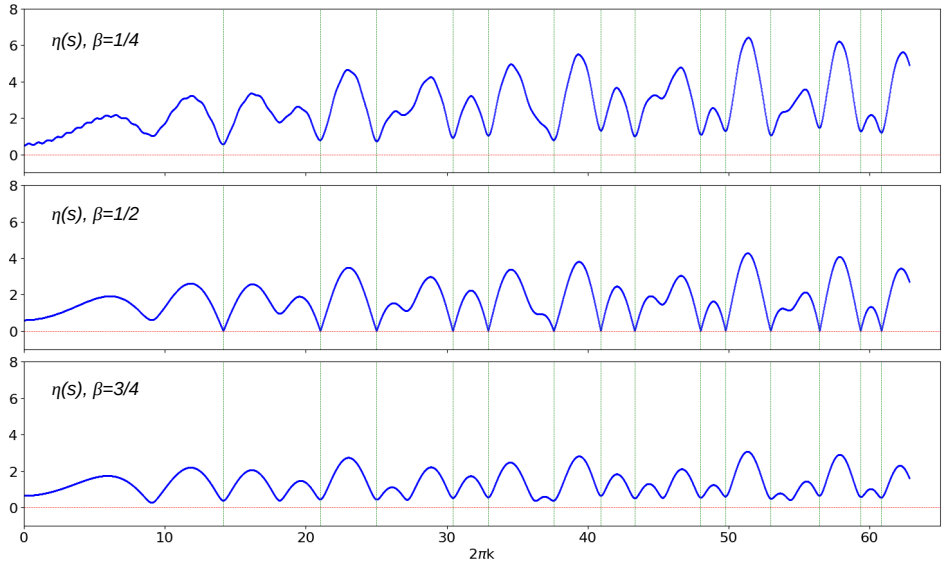


Figure 3. Amplitude spectra of η(s) at β= ¼, ½ and ¾. All non-trivial zeros have a sharp ending on the critical line.

Note 2.1 The non-trivial zero set $\{\gamma_i = 2\pi k_i\}$ represents the characteristic frequencies of the logarithmic series of the natural number system and $\{\lambda_i = 2\pi/\gamma_i\}$ are the wavelengths that represent the intrinsic gaps of the series. The first 5 wavelengths {0.44452122, 0.29888562, 0.25121831, 0.20651474, 0.19077497} and the first 5 gaps of the prime logs {0.40546511, 0.51082562, 0.33647224, 0.45198512, 0.16705408} are on the same scale, where the largest gap 0.51082562 is from the first twin primes 3 and 5.

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